

COMPREHENSIVE 6-WEEK OLIST PRICING TRANSFORMATION PROJECT ROADMAP

PROJECT OVERVIEW

Title: E-Commerce Marketplace Pricing Transformation: A Strategic Approach to Revenue & Customer Lifetime Value Optimization

Business Context:

A Brazilian e-commerce marketplace (Olist) facing:

- High price variance for identical products (seller competition)
- Low customer repeat purchase rates
- Margin pressure from aggressive pricing
- Suboptimal platform commission structure

Objective:

Develop comprehensive pricing strategy combining customer segmentation, marketplace dynamics analysis, and CLV optimization to drive sustainable revenue growth.

WEEK 1 (FEB 17-23): DATA FOUNDATION & BUSINESS CONTEXT

1. Understanding the Tables

Core Transaction Data:

- `olist_orders_dataset.csv` → Order-level activity, time series (order_id, customer_id, order_status, timestamps)
- `olist_order_items_dataset.csv` → Transaction-level pricing analysis (order_id, product_id, seller_id, price, freight_value)
- `olist_products_dataset.csv` → Product characteristics for segmentation (product_id, category, dimensions, photos)

Entity Data:

- `olist_sellers_dataset.csv` → Seller-level analysis (seller_id, location)
- `olist_customers_dataset.csv` → Customer segmentation, geolocation (customer_id, location)

Enrichment Data:

- `olist_order_reviews_dataset.csv` → Customer satisfaction, quality-adjusted pricing (review_id, order_id, review_score)
- `olist_geolocation_dataset.csv` → Geographic segmentation, distance-based logistics
- `olist_payments_dataset.csv` → Payment behavior insights
- `olist_category_name_translation.csv` → English category names

2. Data Cleaning & Merging

Tasks:

- Join tables: orders → order_items → products → customers → sellers → reviews → geolocation
- Convert timestamps to datetime objects
- Handle missing values (freight_value, review_score, product attributes)
- Remove duplicates and outliers
- Create master analytical dataset

Feature Engineering (Initial):

- Order-level: total_spend, num_items, avg_item_price, shipping_cost_ratio
- Time features: day_of_week, month, days_since_last_order
- Geographic: customer_state, seller_state, distance_to_seller

3. Business Problem Framing

Identify Key Issues:

- Calculate price dispersion for identical products across sellers
- Measure customer repeat purchase rate (RFM analysis baseline)
- Analyze margin compression trends
- Assess seller competition intensity by category

Deliverables:

- Jupyter notebook: [01_data_exploration_business_context.ipynb](#)
- Business context memo (2 pages):
 - Market overview
 - Problem statement
 - Analytical approach
 - Expected outcomes

WEEK 2 (FEB 24-MAR 2): CUSTOMER SEGMENTATION & PRICE ELASTICITY

1. Customer Segmentation

RFM Analysis:

- **Recency:** Days since last purchase
- **Frequency:** Number of orders
- **Monetary:** Total spend

Clustering:

- K-means clustering (4-5 segments)
- Hierarchical clustering for validation

Segment Definitions:

1. **High-Value Loyalists** (high F, high M, low price sensitivity)

2. **Bargain Hunters** (high F, low M, high price sensitivity)
3. **Occasional Shoppers** (low F, moderate M, moderate sensitivity)
4. **New Customers** (high R, low F, unknown sensitivity)
5. **At-Risk Churn** (low R, was high F/M)

Additional Segmentation:

- Geographic (state/city clusters)
- Product category preference
- Basket size patterns
- Payment method behavior

2. Price Elasticity Estimation

Approach:

- Log-log regression: $\log(\text{quantity}) \sim \log(\text{price}) + \text{controls}$
- Controls: product category, seller rating, shipping cost, seasonality, reviews
- Estimate by:
 - Product category
 - Customer segment
 - Geographic region

Cross-Price Elasticity:

- Identify substitute products
- Measure competitive effects within categories

Output:

- Elasticity coefficients by segment/category
- Elasticity curves (visualizations)
- Statistical significance and robustness

3. Customer Lifetime Value (CLV) Baseline

Calculate:

$$\text{CLV} = (\text{Avg Order Value}) \times (\text{Purchase Frequency}) \times (\text{Customer Lifespan}) - (\text{Acquisition Cost estimate})$$

By Segment:

- High-value loyalists: \$X CLV
- Bargain hunters: \$Y CLV
- etc.

Deliverables:

- Notebook: `02_segmentation_elasticity_clv.ipynb`

- Visualizations:
 - Customer segment profiles
 - Elasticity curves by category
 - CLV distribution by segment
 - RFM heatmaps
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WEEK 3 (MAR 3-9): SELLER ANALYSIS & MARKETPLACE DYNAMICS

1. Seller Performance Analysis

Metrics:

- High volume sellers (top 20%)
- High ratings (>4.5 stars avg)
- High cancellation/return rates
- Delivery performance
- Price positioning (premium vs discount)

Seller Strategies:

- Identify pricing patterns
- Volume vs margin trade-offs
- Quality positioning

2. Marketplace Dynamics (TWO-SIDED MARKET)

Price Dispersion Analysis:

- Same product, different sellers → measure variance
- Are customers paying premium for high-rated sellers?
- Race-to-bottom dynamics in competitive categories?

Platform Commission Impact:

- Current take rate estimation
- Simulate commission changes:
 - 5% commission → impact on seller pricing & GMV
 - 10% commission → baseline scenario
 - 15% commission → margin optimization
- Show GMV vs platform revenue trade-offs

Buyer-Seller Matching:

- Do high-rated sellers charge premium?
- Customer willingness to pay for quality (reviews impact on price)
- Platform curation strategies

3. Product-Level Recommendations

Price Sensitivity by Category:

- Which categories are most elastic?
- Premium vs commodity positioning

Bundling Opportunities:

- Frequently bought together analysis
- Cross-category bundle potential

Discount Strategy:

- When do discounts increase total revenue?
- Promotional effectiveness by segment

4. Geographic Analysis

Logistics & Pricing:

- High-volume zip codes/states
- Shipping cost sensitivity
- Distance-based pricing adjustments
- Urban vs rural patterns

Deliverables:

- Notebook: `03_seller_marketplace_dynamics.ipynb`
- Insights:
 - Seller performance dashboard
 - Optimal platform take rate
 - Price dispersion analysis
 - Geographic pricing map

WEEK 4 (MAR 10-16): OPTIMIZATION & CLV-DRIVEN STRATEGY

1. Pricing Optimization Framework

Three Objective Functions:**A. Revenue Maximization (Short-term):**

Maximize: Total Revenue
Subject to: Demand elasticity constraints

B. Margin Maximization (Profitability focus):

Maximize: (Revenue – Costs)
Subject to: Minimum volume requirements

C. CLV Maximization (Long-term retention):

Maximize: Customer Lifetime Value
Subject to: Acquisition cost constraints, retention impact

Show trade-offs:

- Revenue max → may sacrifice margin
- Margin max → may lose volume
- CLV max → balanced approach for sustainable growth

2. Segmented Pricing Strategy

By Customer Segment:

- High-value loyalists: Premium pricing, loyalty rewards
- Bargain hunters: Dynamic discounting, volume incentives
- Occasional shoppers: Personalized promotions
- New customers: Acquisition pricing, onboarding offers

By Product Category:

- Electronics: Dynamic pricing, frequent adjustments
- Home & Garden: Seasonal pricing
- Fashion: Clearance strategies

3. Scenario Simulations (Board-Level Metrics)

Simulate:

1. **5% price increase in Electronics category**
 - Impact on: GMV, revenue, units sold, customer churn
2. **10% improvement in customer retention**
 - Impact on: CLV, repeat purchase rate, overall GMV
3. **Platform commission change (10% → 12%)**
 - Impact on: Seller behavior, GMV, platform revenue, margin
4. **Targeted discounting for at-risk customers**
 - Impact on: Churn reduction, CLV improvement, margin trade-off
5. **Premium pricing for high-rated sellers**
 - Impact on: Seller revenue, customer satisfaction, platform GMV

Metrics to Track:

- GMV (Gross Merchandise Value)
- Net revenue (after commission)
- Customer acquisition cost (CAC)
- Customer lifetime value (CLV)
- Customer retention rate
- Average order value (AOV)
- Margin %

4. Bundle & Promotional Strategies

Recommendations:

- Product bundles based on co-purchase analysis
- Cross-category promotions
- Loss leader strategies (low margin products to drive traffic)

Deliverables:

- Notebook: `04_optimization_clv_scenarios.ipynb`
 - Results:
 - Optimal pricing by segment/category
 - Expected impact (12-15% GMV increase, 20% CLV improvement)
 - Scenario simulation results
 - Trade-off analysis
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WEEK 5 (MAR 17-23): EXECUTIVE DECK & DASHBOARD

1. Executive Recommendation Deck (10-15 slides)

Structure:

Slide 1: Executive Summary

- Problem statement
- Key findings (bullet points)
- Expected impact: +12-15% GMV, +20% CLV, +3-5% margin

Slides 2-3: Situation Analysis

- Market context (Brazilian e-commerce, competitive landscape)
- Current challenges:
 - High price variance (X% dispersion)
 - Low retention (Y% repeat rate)
 - Margin pressure (Z% compression)

Slides 4-6: Analysis & Insights

- Customer segmentation (4-5 segments, characteristics)
- Price elasticity findings (category breakdown)
- Marketplace dynamics (seller competition, platform economics)

- CLV analysis (value by segment)

Slides 7-9: Strategic Recommendations

- Segmented pricing strategy (by customer type)
- Platform commission optimization (take rate recommendation)
- Product category strategies (dynamic vs static pricing)
- Seller incentive program (reward high-quality sellers)

Slide 10: Expected Business Impact

- GMV increase: +12-15%
- Customer retention: +20%
- Margin improvement: +3-5%
- CLV increase: +18-22%
- Risk mitigation strategies

Slide 11: Implementation Roadmap

3-Month Pilot Plan:

- **Month 1:**
 - A/B test dynamic pricing in Electronics (high elasticity category)
 - Implement segmented email campaigns
 - Success metrics: GMV change, customer response
- **Month 2:**
 - Roll out personalized pricing to high-value segment
 - Launch seller quality incentive program
 - Success metrics: Retention improvement, seller satisfaction
- **Month 3:**
 - Expand dynamic pricing to 3 additional categories
 - Implement bundle recommendations
 - Success metrics: Overall GMV lift, margin improvement

Slide 12: Success Metrics & Monitoring

- KPIs to track daily/weekly
- Dashboard requirements
- Course correction triggers

Slide 13: Resource Requirements

- Data science team (2 FTE for implementation)
- Pricing platform/software needs
- Change management support
- Timeline: 3 months pilot, 6 months full rollout

Slide 14: Risk Assessment

- Customer backlash to price changes (mitigation: gradual rollout)
- Seller resistance to commission changes (mitigation: incentive program)
- Technical implementation challenges (mitigation: phased approach)

Slide 15: Next Steps

- Approval decision
- Pilot launch timeline
- Stakeholder alignment

2. Power BI Dashboard**Interactive Visualizations:****Overview Tab:**

- GMV trend
- Customer acquisition & retention
- Average order value
- Margin %

Segmentation Tab:

- Customer segment breakdown
- CLV by segment
- Purchase patterns

Pricing Tab:

- Price elasticity heatmap by category
- Elasticity curves (interactive)
- Optimal vs current pricing

Marketplace Tab:

- Seller performance metrics
- Price dispersion visualization
- Platform commission scenario modeling

Geographic Tab:

- State-level performance map
- Shipping cost impact
- Regional pricing recommendations

Filters:

- Date range
- Product category
- Customer segment

- Geographic region

Deliverables:

- [Olist_Pricing_Transformation_Executive_Deck.pdf](#)
 - [Olist_Dashboard.pbix](#) (Power BI file)
 - Slide deck notes/script
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WEEK 6 (MAR 24-31): POLISH, PACKAGE & LAUNCH

1. Code Repository Cleanup

Organize GitHub:

```
ecommerce-pricing-transformation/  
├── README.md (comprehensive overview)  
├── data/  
│   └── README.md (data sources, download instructions)  
├── notebooks/  
│   ├── 01_data_exploration_business_context.ipynb  
│   ├── 02_segmentation_elasticity_clv.ipynb  
│   ├── 03_seller_marketplace_dynamics.ipynb  
│   └── 04_optimization_clv_scenarios.ipynb  
├── src/  
│   ├── data_processing.py (reusable functions)  
│   ├── modeling.py (elasticity, clustering)  
│   └── visualization.py (plotting functions)  
├── outputs/  
│   ├── executive_deck.pdf  
│   ├── technical_report.pdf  
│   └── figures/ (saved visualizations)  
├── requirements.txt  
└── LICENSE
```

Documentation:

- README with:
 - Project overview
 - Business problem
 - Key findings
 - How to reproduce
 - Technologies used
- Comments in all notebooks
- Docstrings for functions
- Data dictionary

2. Technical Report

15-20 pages covering:

1. Introduction (2 pages)

- Business context
- Objectives
- Data overview

2. Methodology (4-5 pages)

- Data preprocessing
- Segmentation approach (RFM + clustering)
- Elasticity estimation (regression specifications)
- CLV calculation
- Optimization framework

3. Results (5-6 pages)

- Customer segments (detailed profiles)
- Elasticity estimates (tables, confidence intervals)
- Marketplace dynamics findings
- CLV analysis
- Optimization results

4. Business Implications (2-3 pages)

- Strategic recommendations
- Implementation considerations
- Expected impact

5. Robustness & Limitations (1-2 pages)

- Sensitivity analyses
- Data limitations
- Model assumptions

6. Appendices

- Additional tables
- Code snippets
- Mathematical derivations

3. LinkedIn Launch Post

Post Structure:

Hook:

"I just completed a comprehensive e-commerce pricing transformation project analyzing 100K+ transactions. Here's what I found about marketplace dynamics and customer lifetime value optimization..."

Context (2-3 paragraphs):

"Using real Brazilian marketplace data, I built an end-to-end pricing strategy combining econometric demand estimation, customer segmentation, and platform economics.

The analysis revealed three critical insights:

1. [Key finding about customer segments]
2. [Key finding about marketplace dynamics]
3. [Key finding about CLV optimization]"

Results (bullets):

"Recommendations showed potential for: • 12-15% GMV increase • 20% customer lifetime value improvement • 3-5% margin expansion • Reduced seller competition intensity"

Technical Approach (1 paragraph):

"I applied discrete choice modeling for elasticity estimation, RFM-based segmentation, and multi-objective optimization for pricing strategy. The project combines PhD-level econometric rigor with business-focused implementation planning."

Call to Action:

"Full project available on GitHub: [link] Executive deck and technical report included.

What's been your experience with marketplace pricing? What challenges have you seen?"

Tags:

- #PricingStrategy
- #RevenueOptimization
- #DataScience
- #Economics
- #Ecommerce

Visual:

- Key visualization from your project (elasticity curves or segment breakdown)

4. Portfolio Website Update

Add Project Page:

- Project title & overview
- Key visualizations (3-4 charts)
- Business impact summary
- Links to GitHub, deck, report
- Technologies used

5. Final Quality Check

Checklist:

-  All notebooks run end-to-end without errors

-  GitHub README is comprehensive
-  Executive deck is polished (no typos, consistent formatting)
-  Power BI dashboard is functional
-  Technical report is professional
-  LinkedIn post is ready
-  All links work

Deliverables (Final):

1. **GitHub Repository** (public, fully documented)
 2. **Executive Deck PDF** (consultant-quality, 10-15 slides)
 3. **Technical Report PDF** (15-20 pages)
 4. **Power BI Dashboard** (.pbix file + screenshots)
 5. **LinkedIn Post** (published, with engagement)
 6. **Portfolio Website** (project page live)
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PROJECT COMPLETION CHECKLIST

By March 31, you will have:

Analytical Depth:

- Customer segmentation (4-5 segments with profiles)
- Price elasticity by category & segment
- Marketplace dynamics analysis (two-sided market)
- CLV calculation & optimization
- Scenario simulations (5+ scenarios)

Business Value:

- Executive recommendation deck (C-suite ready)
- Implementation roadmap (3-month pilot)
- Expected impact quantified (GMV, CLV, margin)
- Risk assessment & mitigation

Technical Excellence:

- Clean, documented code (4-5 notebooks)
- Reproducible analysis
- Professional visualizations
- Power BI dashboard

Consulting Positioning:

- Structured thinking (MECE framework)
- Business-first framing
- Actionable recommendations
- Implementation focus

Portfolio Asset:

- GitHub repository (showcases skills)
 - LinkedIn visibility (thought leadership)
 - Interview talking point (30-min walkthrough ready)
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INTERVIEW NARRATIVES

30-Second Version

"For my portfolio, I analyzed a Brazilian e-commerce marketplace with 100K+ transactions over 2 years. I built an end-to-end pricing transformation focusing on three dimensions: customer lifetime value optimization, marketplace dynamics, and revenue growth.

I estimated demand elasticities, performed RFM segmentation, analyzed two-sided market dynamics, and built a multi-objective pricing optimization framework. The result: 12-15% potential GMV increase with 20% CLV improvement.

I packaged this as an executive deck for C-suite and technical documentation for implementation teams. It demonstrates how economic rigor combines with business strategy to drive measurable impact."

5-Minute Version

Setup (30 sec):

"I wanted to build a project that showcases both analytical depth and business strategy thinking, so I chose e-commerce marketplace pricing—a complex domain with customer behavior, seller competition, and platform economics."

Approach (1 min):

"I structured this like a consulting engagement. Started with business problem framing: the marketplace had high price variance, low retention, and margin pressure. Then built three analytical layers: customer segmentation using RFM and clustering, demand elasticity estimation using discrete choice models, and marketplace dynamics analysis examining seller competition and platform economics."

Key Findings (2 min):

"Three major insights emerged:

First, customers segment into distinct value tiers with very different price sensitivities. High-value loyalists show -0.8 elasticity while bargain hunters are -2.5. This suggests differentiated pricing strategies rather than uniform discounting.

Second, the marketplace shows significant price dispersion—up to 40% variance for identical products. But interestingly, high-rated sellers do command premiums, suggesting quality signaling matters.

Third, customer lifetime value varies dramatically by segment—3x difference between loyalists and bargain hunters. Optimizing for CLV rather than short-term revenue changes the entire pricing

strategy."

Recommendations (1 min):

"I built a three-month pilot roadmap: A/B test dynamic pricing in high-elasticity categories first, implement segment-based personalization for high-value customers, and introduce a seller quality incentive program. The simulations show 12-15% GMV upside with margin improvement, not compression."

Close (30 sec):

"I packaged this as both an executive deck—which I can walk you through—and full technical implementation in Python. It demonstrates how I approach problems: rigorous analysis grounded in economics, but always framed around business impact and implementation reality."

TECHNOLOGIES USED

Programming:

- Python 3.9+
- Jupyter Notebook

Data Analysis:

- pandas
- numpy
- scipy

Modeling:

- statsmodels (elasticity estimation)
- scikit-learn (clustering, segmentation)

Visualization:

- matplotlib
- seaborn
- plotly
- Power BI

Tools:

- Git/GitHub
- VS Code
- PowerPoint/Google Slides

NEXT STEPS

1. **This Week:** Start Week 1 (data exploration)
2. **Send E+H decline:** Wednesday Feb 18

3. **Close F-1:** Thursday Feb 19 (after biometrics)
 4. **Execute roadmap:** Feb 17 - Mar 31
 5. **Launch on LinkedIn:** April 1
 6. **Start job applications:** May (when EAD arrives)
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PROJECT START DATE: February 17, 2026

PROJECT COMPLETION TARGET: March 31, 2026

TOTAL DURATION: 6 weeks